This document will discuss the detailed process of how we did calibration (including program config, input/output files, etc.), **for your information only**, in case you want to redo calibrations. **You can skip this file if you only care about the calibrated results.** The results are in the Calibration/results folder.

Overview

There are three calibration programs used in this folder. It is better (perhaps required) to do them one by one in the following order.

1. Multi-camera calibration using Kalibr toolbox
2. Camera-IMU calibration using Kalibr toolbox
3. LiDAR-camera calibration using ACFR toolbox (i.e. cam_lidar_calibration)

The extrinsic parameters for LiDAR to wheel odometry (origin located at robot base_link), and LiDAR to GPS-RTK are measured and computed from CAD models directly. These two extrinsic parameters do not involve any calibration software.

Multi-camera Calibration

Calibration software: Kalibr
Calibration board: Checkerboard
Input files:

- multi-cam-data-thermal-inverted.bag
- target_checker_board.yaml
- specify the rosbag topic and camera model intended to use for each camera in the command line when running the Kalibr program (currently we use pinhole + radtan for all cameras)

Output files:

- 01-multi-cam-result.yaml
- 01-multi-cam-report.pdf
- 01-multi-cam-report.txt

Notes:

- We use the **interactive_recording.py** script to help record rosbag; whenever the board is placed at a good location, we press the enter key to record this frame, and the output of this script is a rosbag consisting of these frames only.
- Need to run **invert_thermal.py** to process the recorded raw rosbag and invert the intensity of each pixel of thermal images; the multi-cam-data-original.bag is the raw rosbag, and the multi-cam-data-thermal-inverted.bag is the bag processed by this script. (because for the thermal camera, black cells on the checkerboard absorb heat the most, and therefore is shown as the brightest area – white)

IMU-Camera Calibration

Calibration software: Kalibr
Calibration board: April grid
Input files:

- imu-cam-data.bag
- target_april_grid.yaml
- microstrain_gx5.yaml
    - This file was obtained by the allan_variance_ros toolbox, which contains the calibrated IMU intrinsic parameters.
- 01-multi-cam-result.yaml
    - This file is obtained from the multi-camera calibration process.
    - Need to remove other camera info in this file, and only keep cam0 – because we only need cam0 (FLIR blackfly) to be calibrated with the IMU.

Output files:

- 02-imu-cam-result.yaml
- 02-imu-intrinsics.yaml
- 02-imu-cam-report.pdf
- 02-imu-cam-report.txt

LiDAR-Camera Calibration

Calibration software: cam_lidar_calibration
Calibration board: Checkerboard
Input files:

- lidar-cam-data.zip (Zip folder with camera images and lidar point cloud data, as well as some intermediate poses and results)

Output files:

- 03-lidar-cam-result.txt (Just one line of transformation information)

Notes:

- This program requires the camera_info to be published for the camera to be calibrated, so we use **publish_camera_info.py** script to publish **flir_blackfly_cam_info.yaml**, which is copied from the output result of the multi-camera calibration in the first step.

Other Notes:

How to make calibration boards? There are some tricks on this.

- For the april-grid board, we print a poster and use adhesive to stick it to a medium density fiberboard (MDF, a kind of wood board).
- For the checkerboard, we print a poster and glue it to a foam board first. This is because foam is the best material we can find so far to isolate heat (this is to make sure the thermal images do not have blurs on this checkerboard pattern). Then we stick the foam board to a MDF wood board, to make sure the surface is flat. Because normally a foam board does not have a flat surface.

- It is recommended to find a cozy afternoon for calibration, because the sunshine is not too strong nor too weak. Place the checkerboard under direct sunlight and you will get good thermal images.
- During the calibration process, we also cool down the temperature of the checkerboard using a fan after a while (normally 5-10 images being captured), so that the thermal images are kept clear and in high quality in the rosbag.

How to read the multi-camera calibration results?

- This is the default output format of Kalibr, and the result yaml file provides the transformation matrix of the previous camera with respect to the current camera (in other words, how the current camera sees the previous camera), in a chain manner (i.e. they provide T10, T21, T32, T43).
- For example, the matrix under `cam1/T_cn_cnm1` is actually T_cam1_cam0.
- Note that in the table summary we provided in the results folder and our paper, we have inverted these transformation matrices and present them in the format of T_cam0_cam1.

Other scripts?

- The **kalibr_create_target_pdf.py** script is used to generate the 24-inch calibration board pattern, to output both april_grid_24-inch.pdf and checker_board_24-inch.pdf. This script is modified from the Kalibr official code.